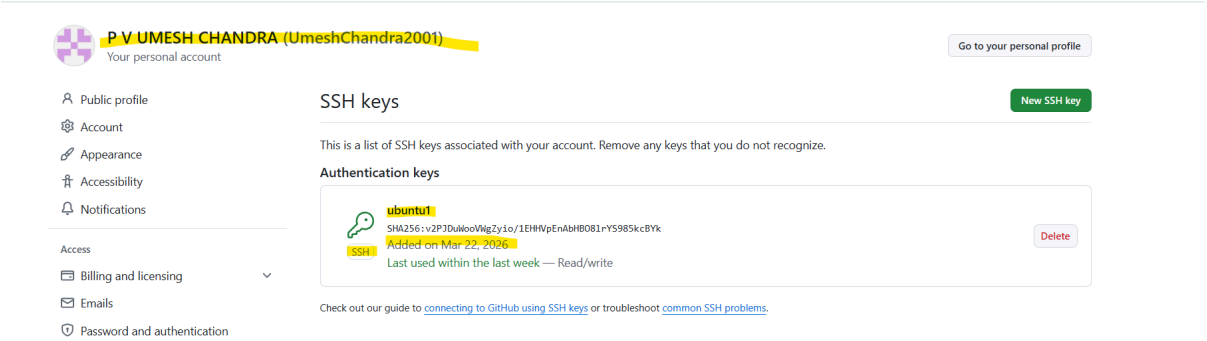
1. Create a new GitHub repository.
    - Clone the repository to your local machine using SSH (generate an SSH key if needed, add the public key to your GitHub account).
    - Create a new branch named after your username (e.g., Tutedude).
    - Add your Flask project files to this branch.
    - Commit the changes and merge the branch into the main branch.

Clone the repository to your local machine using SSH (generate an SSH key if needed, add the public key to your GitHub account

Generated SSH key and added to in my git account:




```
ubuntu@ip-172-31-16-42:~$ git clone git@github.com:UmeshChandra2001/tutedudeassignment4.git
Cloning into 'tutedudeassignment4'...
warning: You appear to have cloned an empty repository.
ubuntu@ip-172-31-16-42:~$ cd tutedudeassignment4
ubuntu@ip-172-31-16-42:~/tutedudeassignment4$ git checkout -b UmeshChandra2001
Switched to a new branch 'UmeshChandra2001'
ubuntu@ip-172-31-16-42:~/tutedudeassignment4$ ls -al
total 12
drwxrwxr-x  3 ubuntu ubuntu 4096 Mar 23 03:36 .
drwxr-x--- 25 ubuntu ubuntu 4096 Mar 23 03:36 ..
drwxrwxr-x  7 ubuntu ubuntu 4096 Mar 23 03:38 .git
ubuntu@ip-172-31-16-42:~/tutedudeassignment4$ git clone git@github.com:UmeshChandra2001/tutedudeassignment4.git
cd tutedudeassignment4
Cloning into 'tutedudeassignment4'...
warning: You appear to have cloned an empty repository.
```

Create a new branch named after your username (e.g., Tutedude).

```
warning: You appear to have cloned an empty repository.
ubuntu@ip-172-31-16-42:~/tutedudeassignment4/tutedudeassignment4$ git checkout -b 1a
Switched to a new branch '1a'
ubuntu@ip-172-31-16-42:~/tutedudeassignment4/tutedudeassignment4$ cp -r /home/ubuntu/TuteDude-Assignments/TD3/1a/* .
```

Add your Flask project files to this branch.


```
warning: You appear to have cloned an empty repository.
ubuntu@ip-172-31-16-42:~/tutedudeassignment4/tutedudeassignment4$ git checkout -b 1a
Switched to a new branch '1a'
ubuntu@ip-172-31-16-42:~/tutedudeassignment4/tutedudeassignment4$ cp -r /home/ubuntu/TuteDude-Assignments/TD3/1a/* .
```


Commit the changes and merge the branch into the main branch.

Now also created new branch called 2a and merged both branched in 1a and 2a into 1a

Merged

2answer file #1

UmeshChandra2001 merged 1 commit into 1a from 2a

None yet

UmeshChandra2001 commented 19 hours ago Owner Author

Mergeing 2a files into 1a

UmeshChandra2001 merged commit 4b969e2 into 1a 19 hours ago Revert

UmeshChandra2001 commented 19 hours ago Owner Author

Merging 2a files into 1a

Pull request successfully merged and closed

You're all set — the 2a branch can be safely deleted.

Delete branch

Add a comment

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

Notifications Customize

Unsubscribe

You're receiving notifications because you modified the open/close state.

1 participant

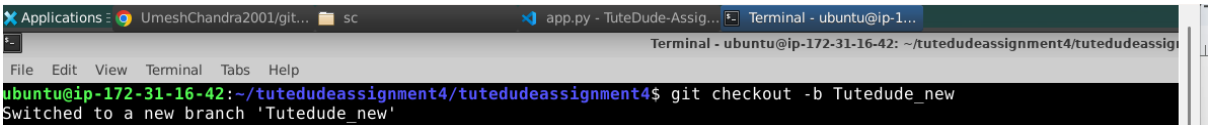
Lock conversation

2. Create a new branch named _new (e.g., Tutedude_new).

- Update the content of the JSON file used for the /api route in this branch.
- Merge the _new branch into the main branch.
- If there are conflicts during the merge, resolve them by accepting the changes from the _new branch.
- Add the resolved changes to the staging area, commit them, and push the updates to the remote repository.

Creating new branch called Tutedude_new

CMD : git checkout -b Tutedude_new



Update Json

CMD : nano data.json

Add Json

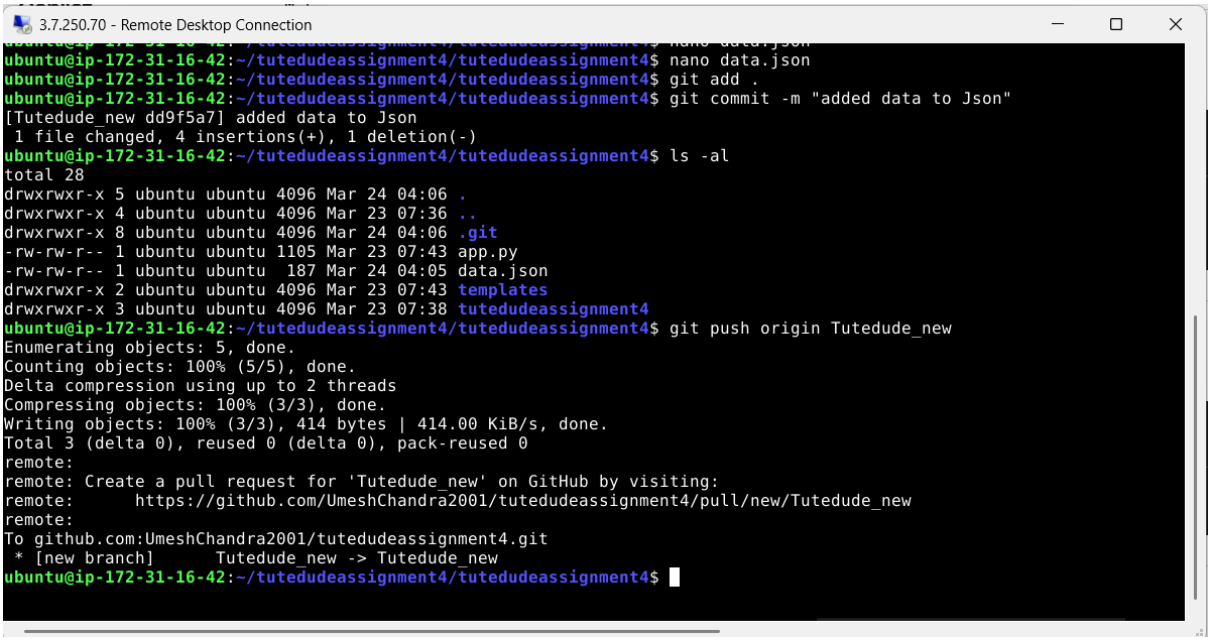
CMD : git add .

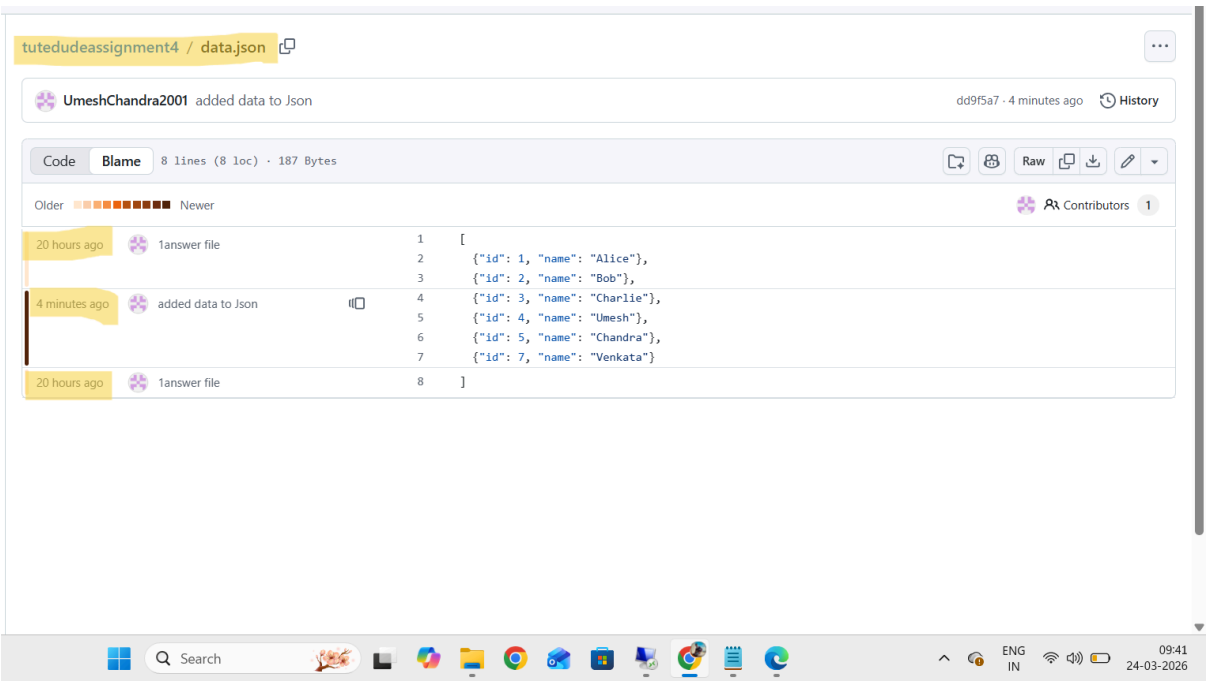
Committing the changes

CMD : git commit -m “added data to JSON”

Push to new branch to git hub

CMD : git push origin Tutedude_new





MERGEING MAIN BRANCH

CMD : git merge Tutedude_new

Push the updated `main` branch

CMD : git push origin main

```
ubuntu@ip-172-31-16-42:~/tutedudeassignment4/tutedudeassignment4$ git checkout main
error: pathspec 'main' did not match any file(s) known to git
ubuntu@ip-172-31-16-42:~/tutedudeassignment4/tutedudeassignment4$ git checkout -b main
Switched to a new branch 'main'
Your branch is based on 'origin/main', but the upstream is gone.
  (use "git branch --unset-upstream" to fixup)
ubuntu@ip-172-31-16-42:~/tutedudeassignment4/tutedudeassignment4$ git checkout main
Already on 'main'
Your branch is based on 'origin/main', but the upstream is gone.
  (use "git branch --unset-upstream" to fixup)
ubuntu@ip-172-31-16-42:~/tutedudeassignment4/tutedudeassignment4$ git merge Tutedude_new
Already up to date.
ubuntu@ip-172-31-16-42:~/tutedudeassignment4/tutedudeassignment4$ git checkout --theirs data.json
Updated 0 paths from the index
ubuntu@ip-172-31-16-42:~/tutedudeassignment4/tutedudeassignment4$ git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'main' on GitHub by visiting:
remote:      https://github.com/UmeshChandra2001/tutedudeassignment4/pull/new/main
remote:
To github.com:UmeshChandra2001/tutedudeassignment4.git
 * [new branch]      main -> main
```

3. Branch Creation:

- Create two branches: master_1 and master_2 from the main branch.
- Feature Development in master_1:
- In the master_1 branch, create a To-Do Page in the frontend.
    - The page should contain a form with the following fields:
        - Item Name
        - Item Description
- Backend API in master_2:
- In the master_2 branch, create a backend route named /submittodoitem.
- This route will:
    - Accept itemName and itemDescription via a POST request.
    - Store these details in a MongoDB database.
- Merging Changes:
- Merge the changes from both master_1 and master_2 into the main branch.

ANS:

Created a repository in GIThub
Open same repository using below comments
CMD : git clone git@github.com:UmeshChandra2001/todo-assignment3.git

Switched to repository using below comments
CMD : cd todo-assignment3/

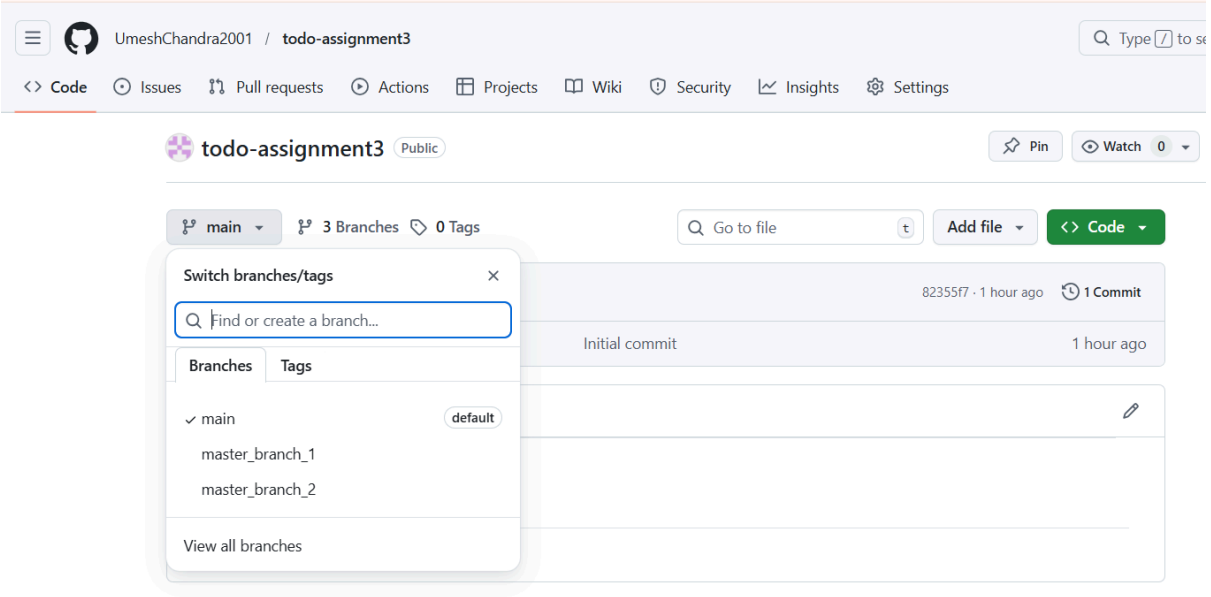
```
ubuntu@ip-172-31-16-42:~$ git clone git@github.com:UmeshChandra2001/todo-assignment3.git
Cloning into 'todo-assignment3'...
Enter passphrase for key '/home/ubuntu/.ssh/id_ed25519':
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
ubuntu@ip-172-31-16-42:~$ cd todo-assignment3/
```

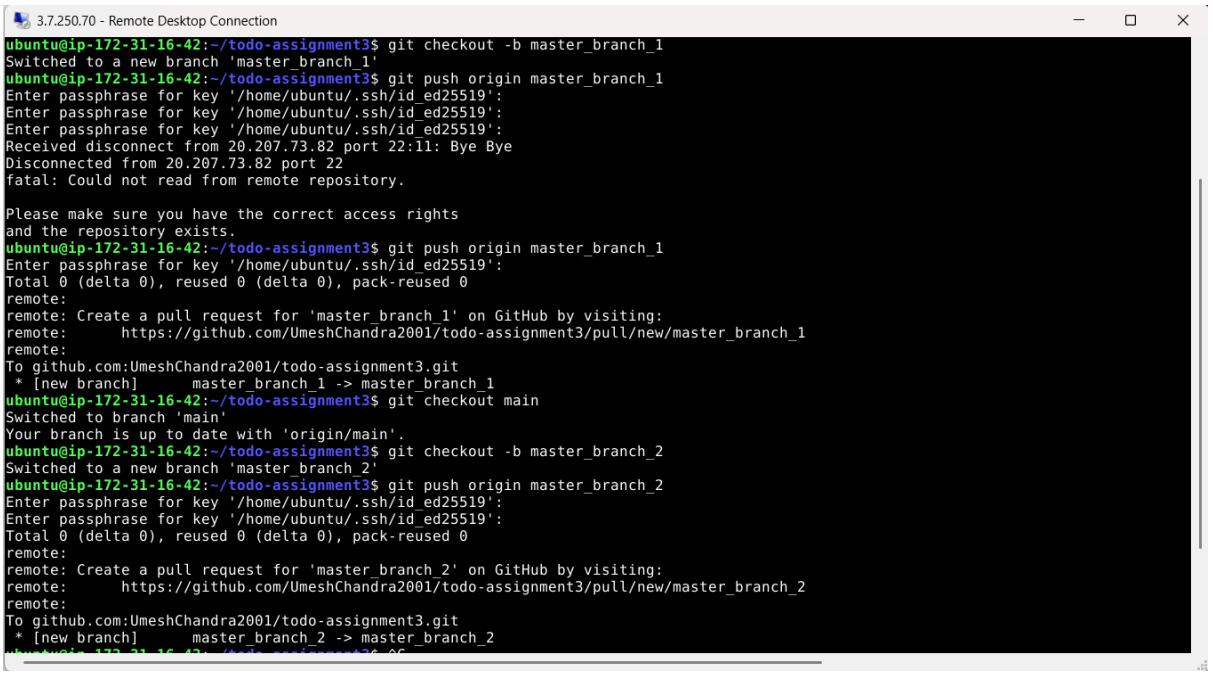
Now Created two new branches using below commands

CREATING BRANCH : git checkout -b master_branch_1
PUSH BRANCH TO GIT : git push origin master_branch_1

Getting out branch to main branch : git checkout main
CREATING BRANCH : git checkout -b master_branch_2

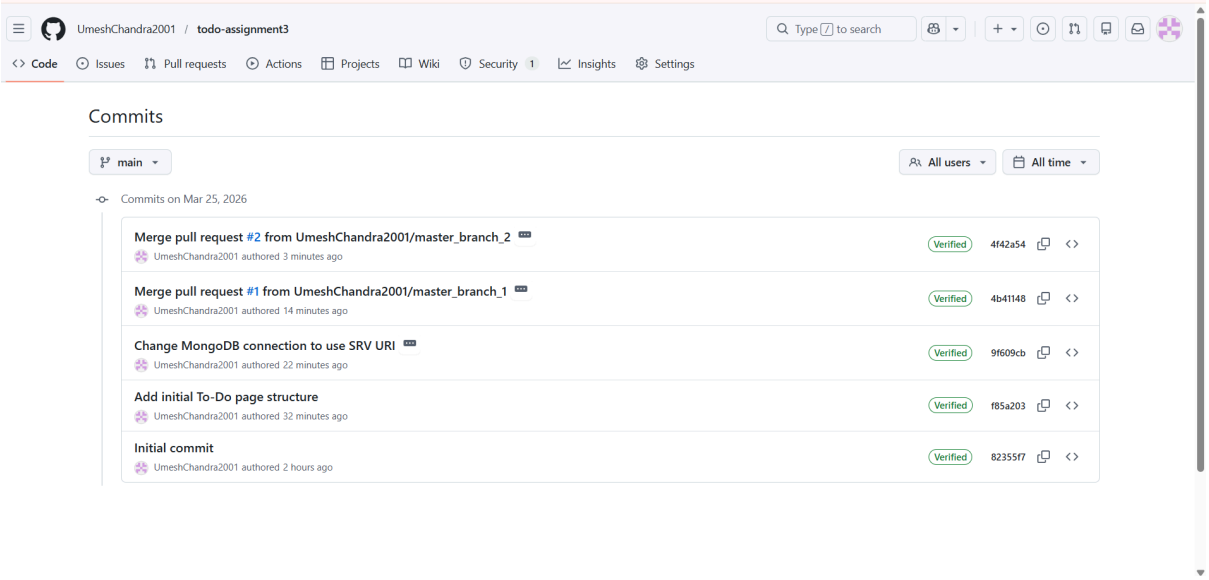
PUSH BRANCH TO GIT : git push origin master_branch_2





Switching to master_branch_1
CMD : git checkout master_branch_1

MERGING OF TWO BRANCHES



CODE are stored in path : https://github.com/UmeshChandra2001/todo-assignment3

4. Enhancing the To-Do Form in master_1:
● In the master_1 branch, add the following fields to the To-Do form:
    ○ Item ID
    ○ Item UUID
    ○ Item Hash
● Committing in Sequence:
● Add and commit each field separately in the following order:
    ○ First commit: Add Item ID field.
    ○ Second commit: Add Item UUID field.
    ○ Third commit: Add Item Hash field.
● Merging to main:
● Merge the master_1 branch into the main branch.
● Git Reset and Commit Deletion:
● In the main branch, use Git Reset to roll back to the commit where only the Item ID field was added.
● Use git reset --soft to ensure changes remain staged.
● Re-commit this state to the main branch.
● Merge this updated state to the main branch.
● Rebasing Changes:
● Rebase the updated changes in the main branch to the master_1 branch.
Clarification:
    ○ During rebasing, preserve individual commits to maintain the commit history for each change (i.e., do not squash commits).
    ○ Use git rebase main master_1 to integrate changes from the main branch back into the master_1 branch.

ANS:

Creating new repository todo-assigment4

Creating new branch and changing:
CMD : git checkout -b master_branch_1

First commit: Add Item ID

git add todo.html
git commit -m "Added Item ID field"

```
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git checkout -b master_branch_1
Switched to a new branch 'master_branch_1'
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git add^C
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git add todo.html
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git commit -m "Added Item ID field"
[master_branch_1 91af86f] Added Item ID field
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 todo.html
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git add todo.html
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git commit -m "Added Item UUID field"
On branch master_branch_1
nothing to commit, working tree clean
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git push origin master_branch_1
Enter passphrase for key '/home/ubuntu/.ssh/id_ed25519':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 301 bytes | 301.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'master_branch_1' on GitHub by visiting:
remote:      https://github.com/UmeshChandra2001/todo-assignment4/pull/new/master_branch_1
remote:
To github.com:UmeshChandra2001/todo-assignment4.git
 * [new branch]      master_branch_1 -> master_branch_1
```

Second commit: Add Item UUID

git add todo.html
git commit -m "Added Item UUID field"

```
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git add todo.html
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git commit -m "Added Item UUID field"
[master_branch_1 3c18534] Added Item UUID field
 1 file changed, 23 insertions(+)
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git push origin master_branch_1
Enter passphrase for key '/home/ubuntu/.ssh/id_ed25519':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 520 bytes | 520.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:UmeshChandra2001/todo-assignment4.git
   91af86f..3c18534  master_branch_1 -> master_branch_1
```

Third commit: Add Item Hash

git add todo.html
git commit -m "Added Item Hash field"

```
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git add todo.html
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git commit -m "Added Item Hash field"
[master_branch_1 3af9230] Added Item Hash field
 1 file changed, 3 insertions(+)
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git push origin master_branch_1
Enter passphrase for key '/home/ubuntu/.ssh/id_ed25519':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 369 bytes | 369.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:UmeshChandra2001/todo-assignment4.git
   3c18534..3af9230  master_branch_1 -> master_branch_1
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$
```

```
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git log
commit 3af92307ffa67fb8f0ac3712f79e62cac0428f84 (HEAD -> main, origin/master_branch_1, origin/main, master_branch_1)
Author: Umesh Chandra <potthuruvenkataumeshchandra@gmail.com>
Date:   Wed Mar 25 08:15:04 2026 +0000

    Added Item Hash field

commit 3c1853430e59e81d8e28919ab33928b9912e9fb9
Author: Umesh Chandra <potthuruvenkataumeshchandra@gmail.com>
Date:   Wed Mar 25 08:07:30 2026 +0000

    Added Item UUID field

commit 91af86f704e07abd37469c88fb01253f338792a1
Author: Umesh Chandra <potthuruvenkataumeshchandra@gmail.com>
Date:   Wed Mar 25 08:04:03 2026 +0000

    Added Item ID field

commit d0f83d1b8eba501bd971427c6f64f664b9bbf0b4
Author: Umesh Chandra <potthuruvenkataumeshchandra@gmail.com>
Date:   Wed Mar 25 07:48:24 2026 +0000

    Initial commit
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$
```

todo.html ./ | todo.html todo-assignment4 | app.py

todo-assignment4 > todo.html > html > body > form#todoForm

You, 5 minutes ago | 1 author (You)

```html
<!DOCTYPE html>
<html>
<head>
  <title>To-Do Page</title>
</head>
<body>
  <h1>Create a To-Do Item</h1>
  <form id="todoForm">
    <!-- These comment 1 -->
    <label>Item ID:</label>
    <input type="text" name="itemId" required><br><br>

    <!-- These comment 2-->
    <label>Item UUID:</label>
    <input type="text" name="itemUUID" required><br><br>

    <!-- These comment 3-->      You, 5 minutes ago • Uncommitted changes
    <label>Item Hash:</label>
    <input type="text" name="itemHash" required><br><br>


    <button type="submit">Submit</button>

  </form>
</body>
</html>
```

Merge `master_branch_1` into `main`:

```
git checkout main
git merge master_branch_1
git push origin main
```

```
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git merge master_branch_1
Updating d0f83d1..3af9230
Fast-forward
 todo.html | 26 ++++++++++++++++++++++++++
 1 file changed, 26 insertions(+)
 create mode 100644 todo.html
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git push origin main
Enter passphrase for key '/home/ubuntu/.ssh/id_ed25519':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:UmeshChandra2001/todo-assignment4.git
   d0f83d1..3af9230  main -> main
```

git log --oneline
git reset --soft 3af9230
git commit -m "Re-commit state with only Item ID field"
git push origin main --force

```
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git log --oneline
3af9230 (HEAD -> main, origin/master_branch_1, origin/main, master_branch_1) Added Item Hash field
3c18534 Added Item UUID field
91af86f Added Item ID field
d0f83d1 Initial commit
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git reset --soft 3af9230
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git commit -m "Re-commit state with only Item ID field"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git add .
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git commit -m "Re-commit state with only Item ID field"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git push origin main --force
Enter passphrase for key '/home/ubuntu/.ssh/id_ed25519':
Everything up-to-date
```

git checkout master_branch_1
git rebase main
git add todo.html
git rebase --continue
git push origin master_branch_1 --force

```
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git checkout master_branch_1
Already on 'master_branch_1'
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git rebase main
Current branch master_branch_1 is up to date.
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git add todo.html
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git rebase --continue
fatal: No rebase in progress?
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$ git push origin master_branch_1 --force
Enter passphrase for key '/home/ubuntu/.ssh/id_ed25519':
Everything up-to-date
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/todo-assignment4$
```